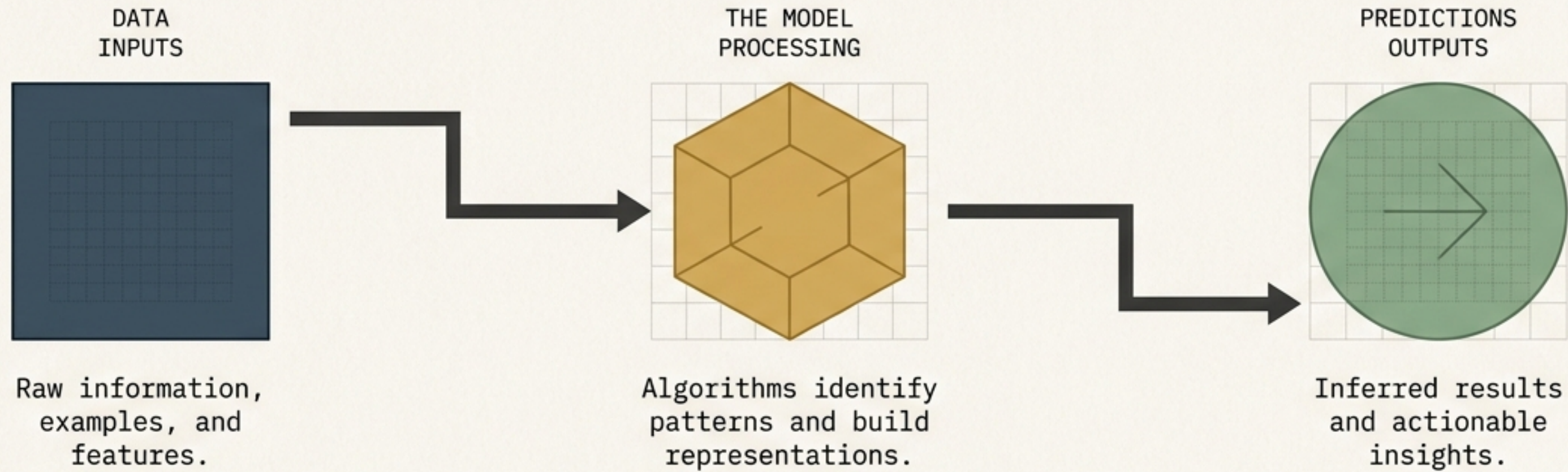


How Computers Learn

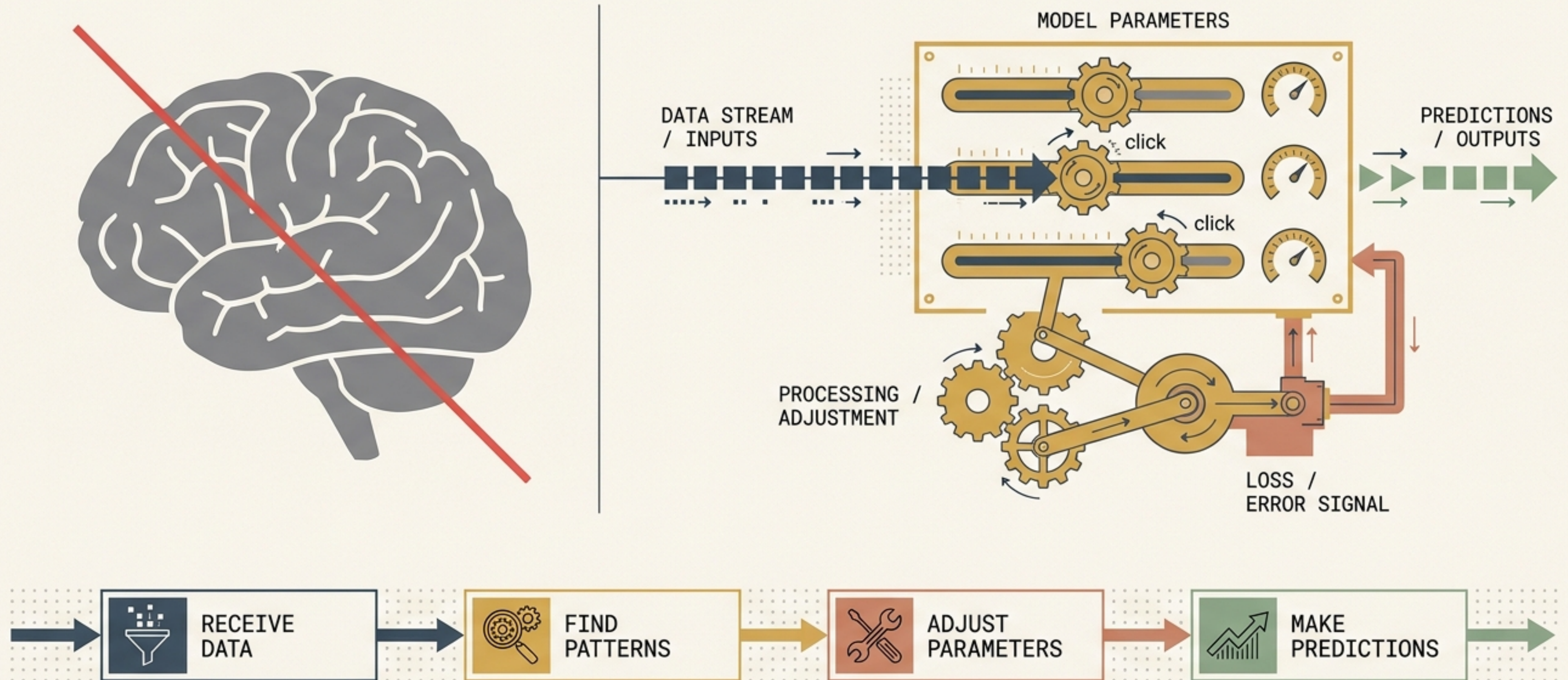
From Data → Patterns → Models → Predictions



If computers don't have a brain...
how are they learning?

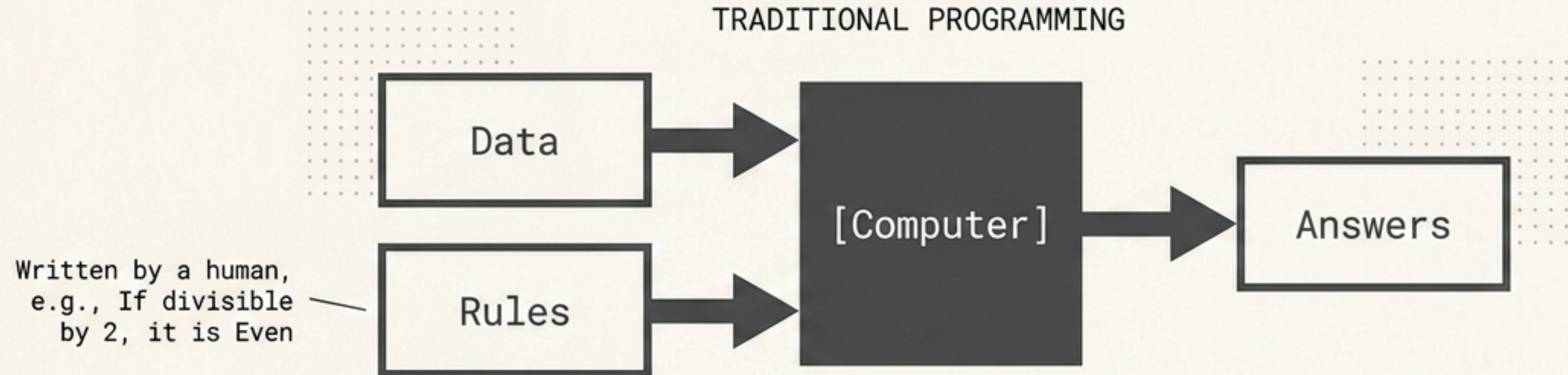
Computers Do Not Become Conscious

When a computer “learns,” it is simply adjusting internal parameters based on experience.

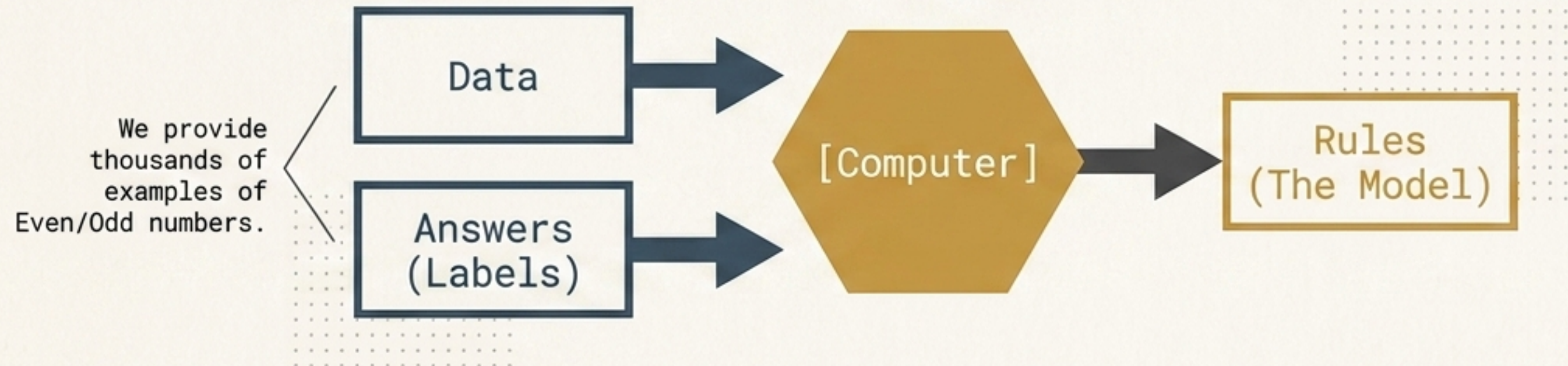


The Paradigm Shift

TRADITIONAL PROGRAMMING



MACHINE LEARNING



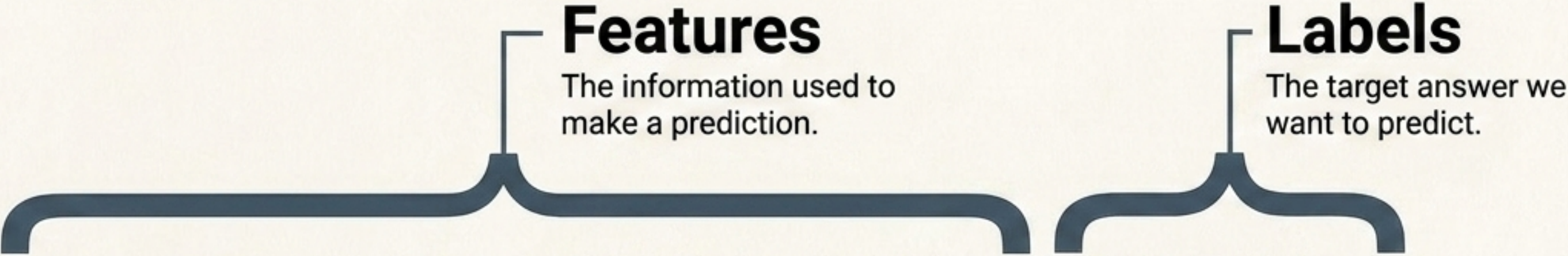
Deconstructing the Raw Material

Features

The information used to make a prediction.

Labels

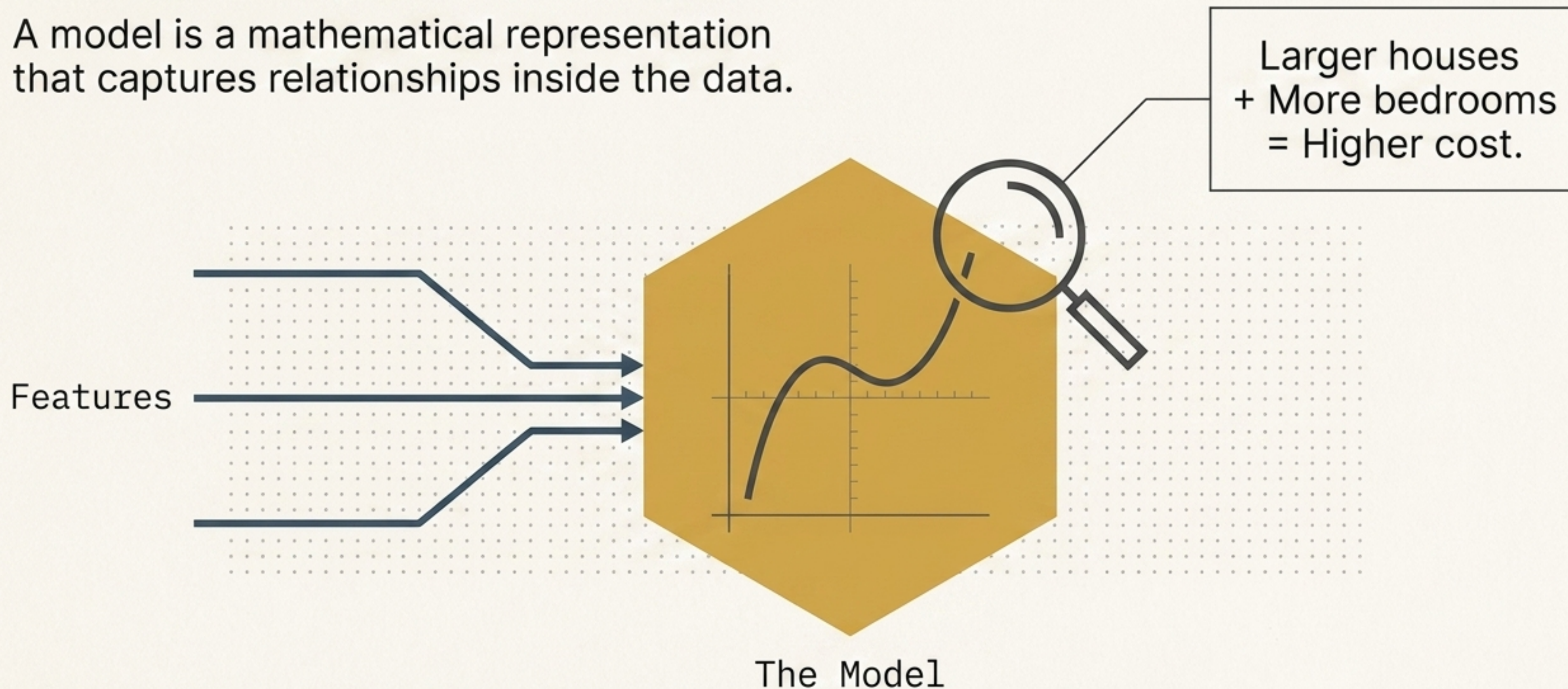
The target answer we want to predict.



Size	Bedrooms	Location	Price
800 sq ft	1	Pune	₹40L
1200 sq ft	2	Pune	₹60L
1800 sq ft	3	Pune	₹90L

The Model is a Learned Pattern

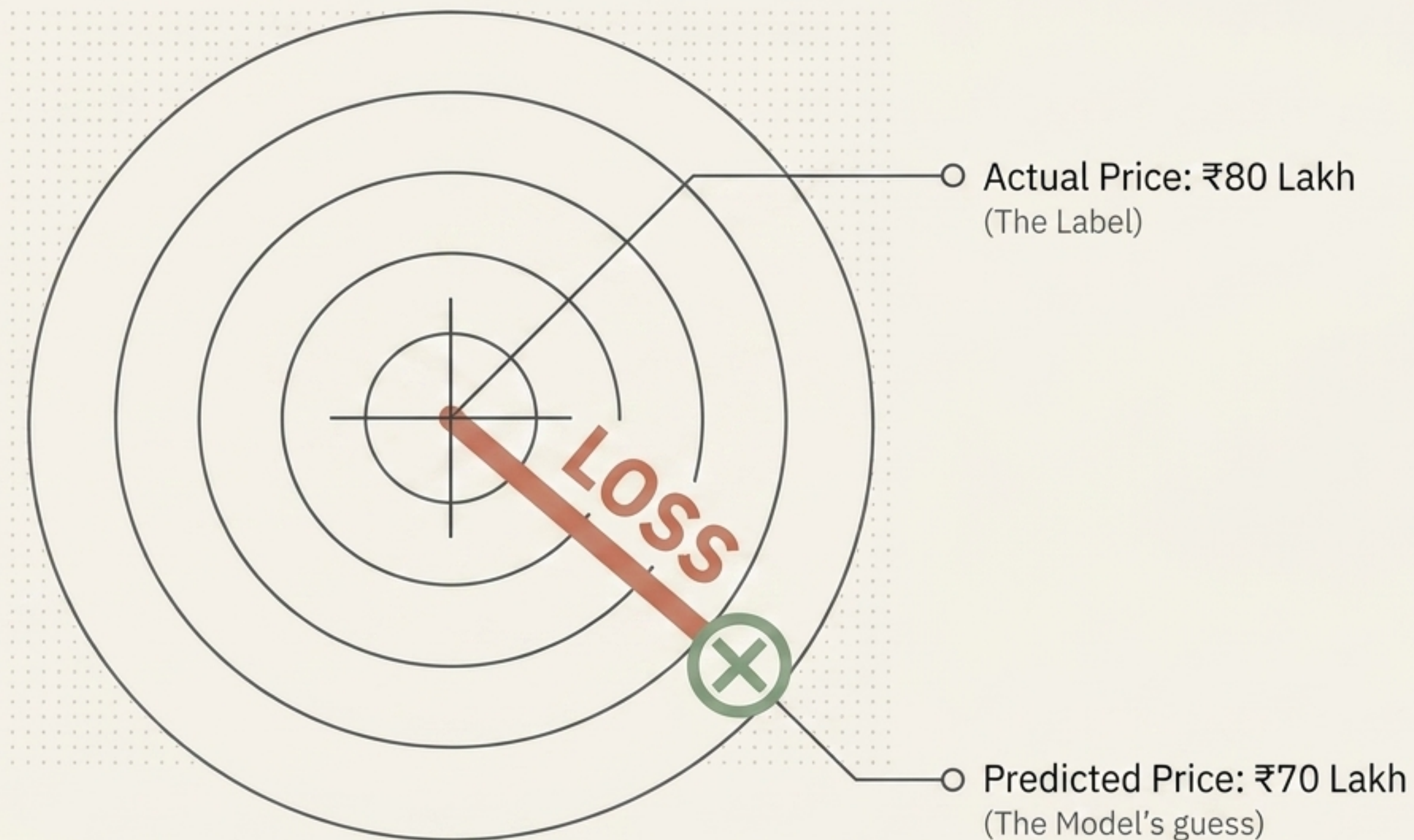
A model is a mathematical representation that captures relationships inside the data.



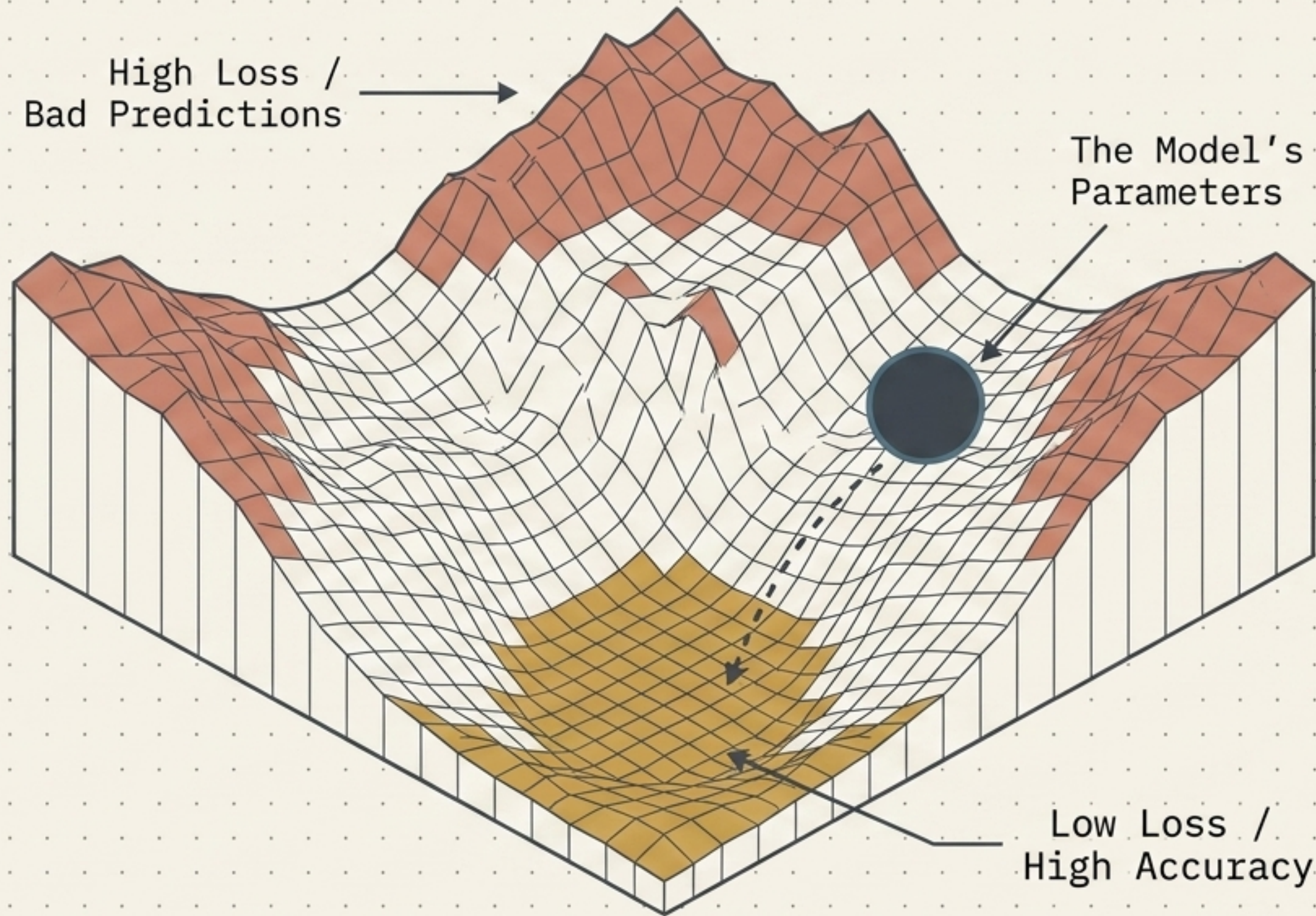
Real models learn patterns far more complex than human-written rules.

Measuring the Error Gap

A loss function converts the prediction error into a numerical value. Lower loss means the model is closer to the target.



Optimization: Rolling Down the Hill

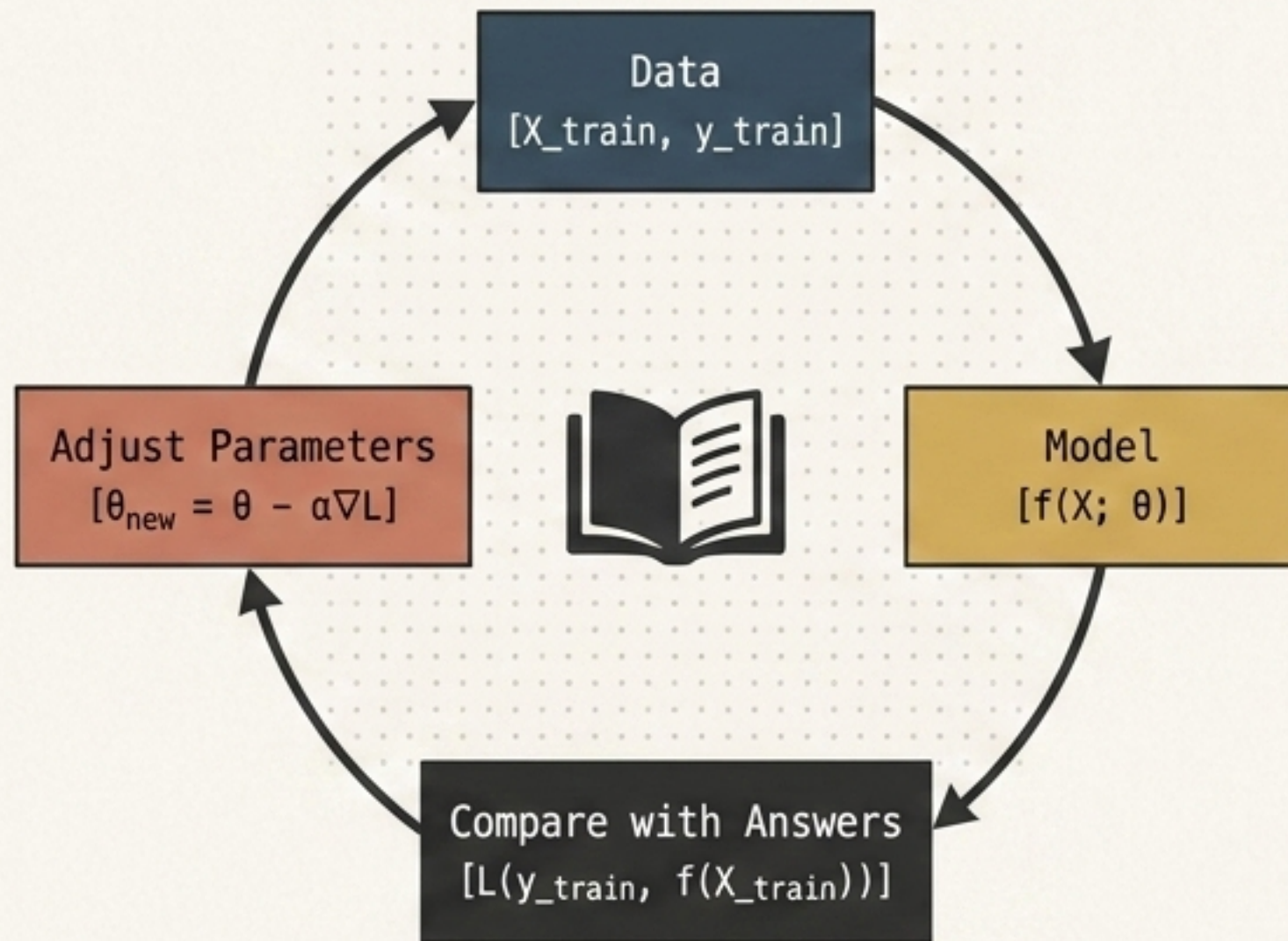


1. Check the Loss.
2. Calculate the slope (Gradient).
3. Update parameters to move downhill.
4. Repeat until error is minimized.

Two Distinct States of Existence

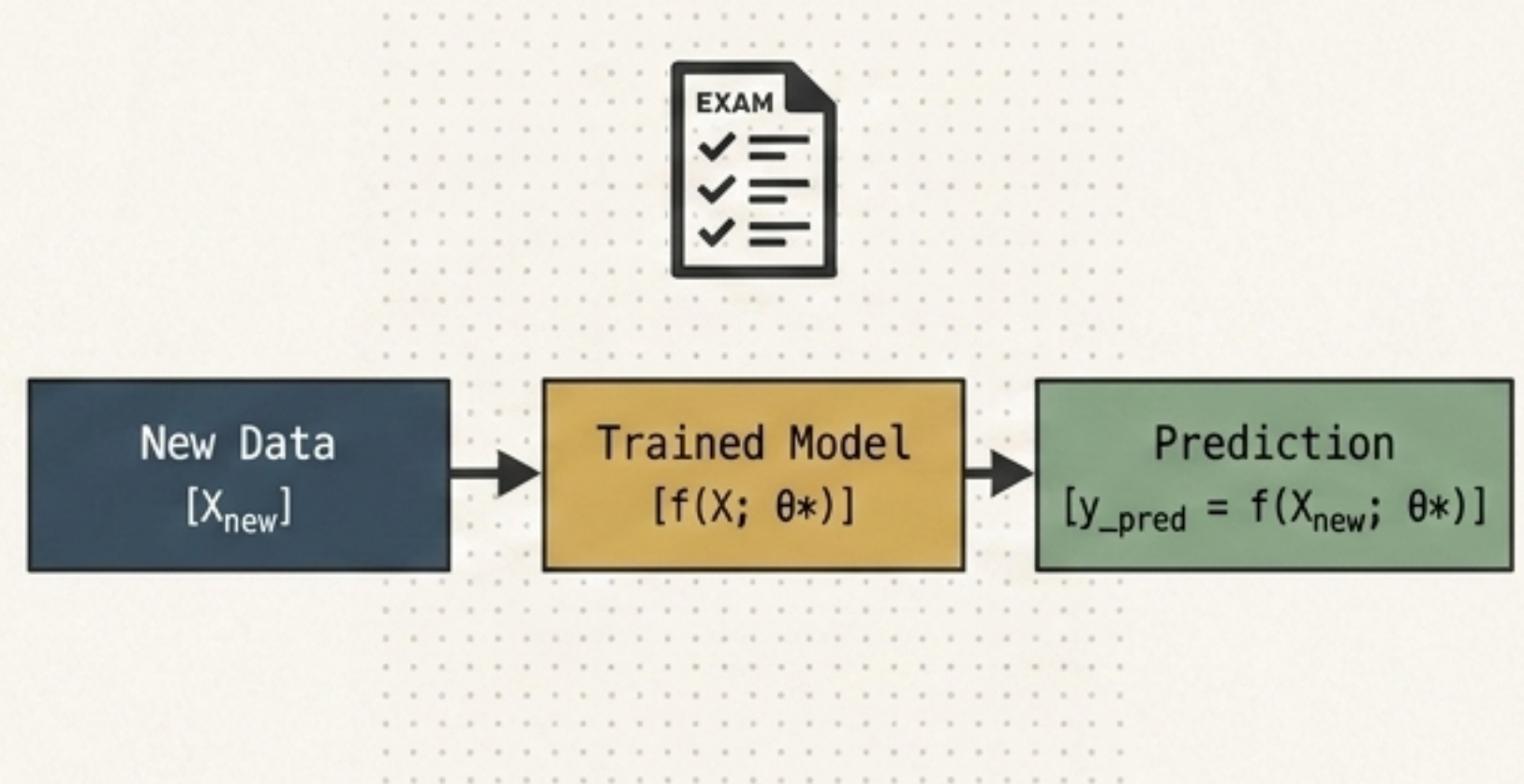
Training = Studying

The model adjusts its internal parameters using an answer key.

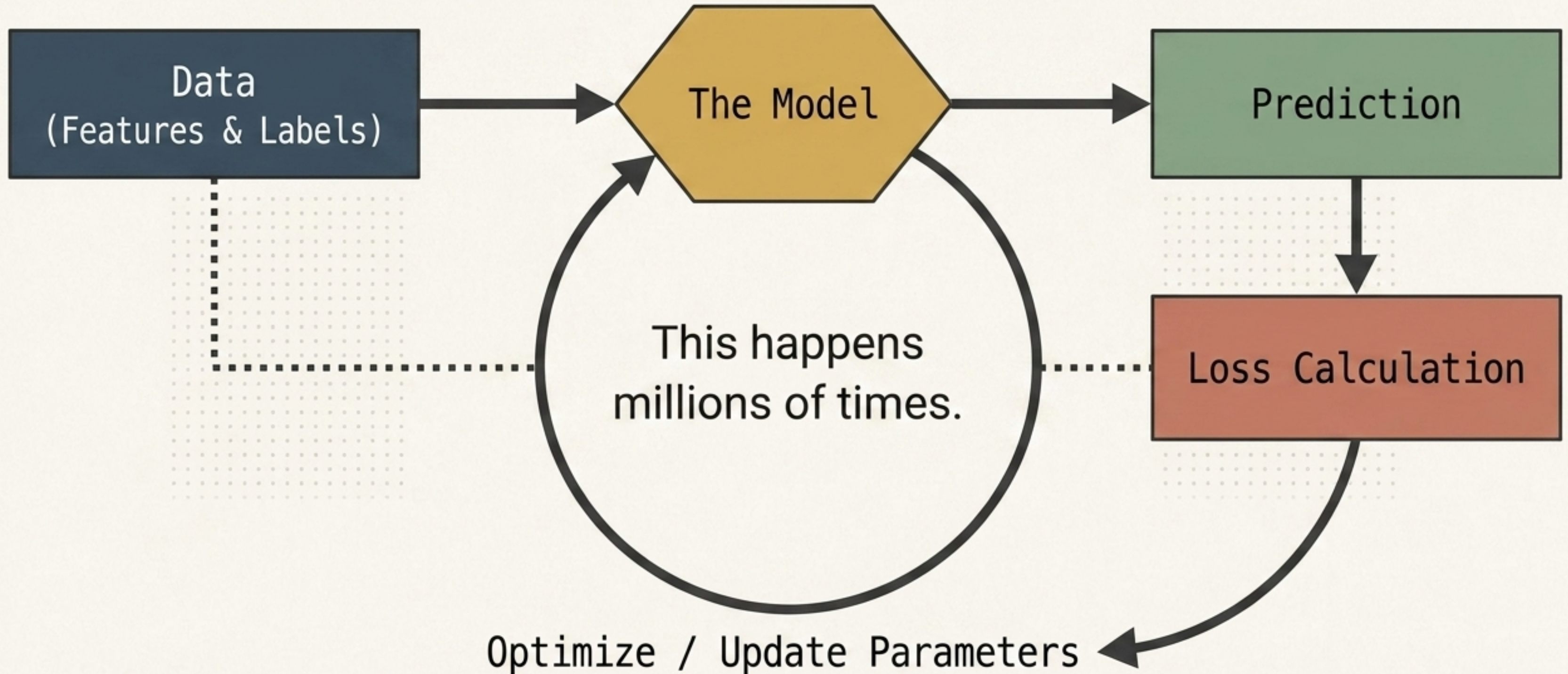


Inference = Taking the Exam

The trained model operates in the real world with no answer key.



The Complete Learning Loop



Machine Learning is Not Magic

It is an optimization process where parameters are adjusted to improve a defined objective.

